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328312 (14)

BE (3rd Semester)

Examination, April-May, 2013

Branch : EEE, Et & T

NUMERICAL ANALYSIS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : In each question part (a) is compulsory, it carries two marks. Attempt any two parts from (b), (c) and (d) in each question, it carry seven marks for each.

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Q. 1. (a) (i) If $f(a) > 0$ and $f(b) > 0$, we can find a value of x where $a < x < b$ so that $f(x) = 0$ [True or false]. 1

(ii) Gauss-Jordan method is an _____ method. [Direct or iterative] 1

(b) Using Bisection method, find the negative root of $x^3 - 4x + 9 = 0$. 7

(c) Use Newton's Raphson method to find a root of the equation : 7

$x \sin x + \cos x = 0$ which is near $x = \pi$ to 3 places of decimals.

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(d) Perform two iterations by Bairstow's method to extract a quadratic factor $x^2 + px + q$ from the polynomial $x^3 + x^2 - x + 2 = 0$ using $p = -.9$, $q = .9$ as initial approximation. 7

Q. 2. (a) Explain briefly Gauss elimination method of Gauss Jordan method. 2

(b) Apply Gauss Jordan method to solve : 7

$x + 2y + z = 8,$

$2x + 3y + 4z = 20,$

$4x + 3y + 2z = 16$

(c) Solve by Crout's method : 7

$2x + 3y + z = 9$

$x + 2y + 3z = 6$

$3x + y + 2z = 8$

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(d) Solve the system of equations by relaxation method : 7

$12x + y + z = 31$

$2x + 8y - z = 24$

$3x + 4y + 10z = 58$

Q. 3. (a) P.T. 2

$E = 1 + \Delta$

(b) Given that : 7

x :	1.0	1.1	1.2	1.3	1.4	1.5	1.6
y :	7.989	8.403	8.781	9.129	9.451	9.75	10.031

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$

(i) at $x = 1.1$

(ii) at $x = 1.6$

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(c) Find an approximate value of $\log_e 5$ by calculating to 4 decimal places by Simpson's $\frac{1}{3}$rd rule. 7

$$\int_0^5 \frac{dx}{4x+5}$$

dividing the range into 10 equal parts.

(d) The velocity v [km/min] of a moped which starts from rest is given at fixed intervals of time 't' (min) as follows : 7

t :	2	4	6	8	10	12	14	16	18	20
v :	10	18	25	29	32	20	11	5	2	0

Estimate approx. the distance covered.

Q. 4. (a) Write the predictor and corrector formula for Adams-Bashforth method. 2.

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(b) Solve : 7

$$\frac{dy}{dx} = x + y^2, \; y(0) = 1,$$

using Taylor's series method and compute y(.1) & y(.2).

(c) Given that : 7

$$\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$$

and y = 1 at x = 0, find y for x = .4, x = .2

(d) Use Milne's predictor-corrector method to obtain the solution of the equation : 7

$$\frac{dy}{dx} = x - y^2 \text{ at } x = .8$$

Given that :

y(0) = .000, y(.2) = .0200

y(.4) = .0795, y(.6) = .1762

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Q. 5. (a) Write normal equations for the curve : 2

$y = ab^x$

(b) Fit a second degree parabola to the following data : 7

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
	1.1	1.3	1.6	2.0	2.7	3.4	4.1

(c) The pressure and volume of a gas are related by the equation : 7

$pv^r = k$, r and k being constant. Fit this equation to the following set of observations.

p[kg/cm^2]	.5	1.0	1.5	2.0	2.5	3.0
v[litres]	1.62	1.00	.75	.62	.52	.46

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(d) Fit the curve : 7

$y = ae^{bx}$

to the following data

x :	1	5	7	9	12
y :	10	15	12	15	21

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